Week 8, Lecture 1 WOD: Answer Key

How Much Did the Prior Matter?

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. 2a is open-ended and gets discussion points. All three panels are Beta-binomial (the setup from Week 7 Wednesday) with the same data, 21 of 30 collared tortoises surviving, so the shared likelihood peaks at 0.70. Students don't see the data or any of these numbers.

Analyst	Prior	Prior worth (tortoises)	Prior mean	Posterior mean	95% credible interval
A	Beta(2, 2)	4	0.50	0.68	0.51 to 0.82
B	Beta(46, 4)	50	0.92	0.84	0.75 to 0.91
C	Beta(10, 10)	20	0.50	0.62	0.48 to 0.75

A Beta(α, β) prior is "worth" about $\alpha + \beta$ animals, against 30 collared tortoises in the data.

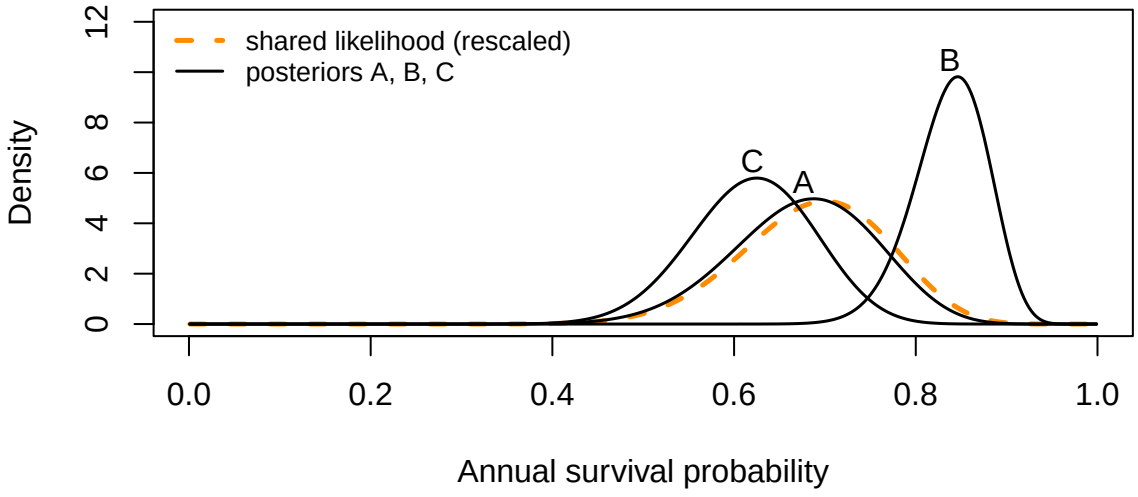
Use the debrief to lead into the germination example: the supplier prior there is worth 200 seeds against 40 seeds of data, a stronger prior relative to the data than Analyst B's.

Part 1: Find the likelihood

1a.

> Sketch the shared likelihood on the bottom panel (the height doesn't matter; get its peak and width roughly right). Which analyst's panel did you rely on most, and why?

Answer. A curve peaking near 0.70 and about as wide as Analyst A's posterior (roughly 0.5 to 0.85). Rely on Analyst A: with a nearly flat prior, A's posterior is almost the likelihood itself.



Why. The posterior is a compromise between prior and likelihood. When the prior is nearly flat it barely pulls, so the posterior is close to the rescaled likelihood. The other two panels check the sketch: B's posterior sits between its prior (near 0.92) and 0.70, and C's sits between its prior (0.50) and 0.70.

Watch for. Averaging the three posteriors (that bakes in B's and C's priors). Drawing the likelihood at B's posterior peak because B's curve is the tallest and narrowest. Students who use B or C should still notice the likelihood must sit on the far side of the posterior from that analyst's prior.

1b.

> Rank the three analysts' results from most to least dependent on their choice of prior. Give a one-line reason for each.

Answer. B (most), then C, then A (least).

- **B:** posterior pulled far from the likelihood toward 0.92; its prior is worth more than the data (50 vs. 30 tortoises).
- **C:** posterior pulled partway toward 0.50; a moderate prior (worth 20).
- **A:** posterior nearly matches the likelihood; a weak prior (worth 4).

Why. Every analyst used the same data, so the differences among posteriors come entirely from the priors. How far each posterior sits from the likelihood measures how much its prior mattered.

Watch for. Ranking by posterior width. B's posterior is the narrowest, but that narrowness comes from the prior, not from more data (all three had the same data).

Part 2: Which number?

2a.

> The three posterior means for annual survival are 0.68 (A), 0.84 (B), and 0.62 (C). A manager needs one survival rate for a population model. What would you tell them, and what would you report?

Discussion points.

- The conclusion is **sensitive to the prior**: the same data give posterior means from 0.62 to 0.84. For a long-lived species, that spread is the difference between a growing and a declining population. This is a prior sensitivity analysis, and the right response is to report it, not hide it.
- **Which prior is defensible is an ecological question.** Is a well-protected population a good guide to this site? Was this year a drought year? If neither earlier population resembles this site, Analyst A's weak prior (letting the data speak) is the safest single number.
- **What to report:** the posterior under the most defensible prior, plus the other two, and the plain statement that 30 tortoises aren't enough to make the prior irrelevant. More data would shrink the disagreement.

Watch for. "Average the three." Averaging posteriors doesn't fix an unjustified prior. "Use B, it's the most certain." B's narrow interval comes from its prior. Both are worth raising with the room.